Unit 4 Kitchen Electrical — Internal Working Plan R1

2026-11-02 · English edition (Spanish edition is a separate file) · **DRAFT for Tisa + licensed electrician review.** Not a permit set, purchase list or rough-in release.

Bottom line

- **Working drawings KE-01 to KE-05 Rev D** show every box and route, with the room to scale from Magicplan (kitchen 147 × 126 in, ceiling 121 1/2 in).
- **Countertop receptacle layout passes code** as drawn (E1/E2, S1/S2, W1/W2). No extra counter receptacles needed.
- **Every kitchen circuit needs AFCI, and every kitchen receptacle needs GFCI** (2025 CEC). Simplest way: dual-function (AFCI+GFCI) breakers at the D04 subpanel for the 7 receptacle circuits; AFCI for the hood and lights.
- **9 kitchen circuits, unchanged from the E20 register.** No added panel spaces. One optional saving (hood on lighting circuit) depends on the Zephyr label.
- **Permit:** SF only issues homeowner electrical permits for single-family dwellings. This is a multi-unit building, so a **licensed electrical contractor pulls the permit, and only that contractor or their employees may do the wiring** (SF Bldg Code 89.120(E), 89.122(A)). The current sprint wording that "crew can execute wiring" needs to change.
- **4 conflicts to settle before marking boxes** — see Open items K-E01 to K-E04.

Code basis

Code	Edition	Applies to
California Electrical Code, Title 24 Part 3	2025 (NEC 2023 base), effective 1 Jan 2026	Circuits, protection, receptacle layout
San Francisco Electrical Code amendments	2025	300.4(I), 230.71, 334 NM-cable limits
SF Building Code Ch. 1A / Sec. 89	current	Who may pull and perform electrical work
California Energy Code, Title 24 Part 6	2025	Lighting efficacy and controls (§150.0(k))

The edition is set by the permit application date. No Unit 4 electrical permit was found in the project records; if one was filed before 1 Jan 2026, the 2022 cycle applies and this check must be re-run.

Circuit schedule (D04 subpanel, Eaton BRP20L125G)

ID	Serves	Breaker	Romex (Cu NM-B)	Protection	Device	Why
SA-A	East counter E1, E2	20 A	12/2	AFCI+GFCI (DF)	20 A TR duplex	210.11(C)(1), 210.52(B)(3), 210.8(A)(6), 210.12(A)
SA-B	South counter S1, S2	20 A	12/2	DF	20 A TR duplex	same
SA-C	West counter W1, W2	20 A	12/2	DF	20 A TR duplex	same
REF	Haier refrigerator	20 A	12/2	DF	single TR receptacle	210.52(B)(1) Ex. 2; Haier: own outlet, 15 or 20 A
DW	Bosch dishwasher	15 A	14/2 min.	DF	receptacle in sink base, or hardwire to Bosch box	Bosch: dedicated 15 A, #14 Cu 75 °C; 210.8(D)
GD	Disposal (switched)	15 A*	14/2 + switch loop	DF	switched receptacle in sink base; motor-rated switch	210.8(A)(6); 422.31
RG	Viking gas range	15 A	14/2	AFCI + GFCI — method open (K-E03)	single TR receptacle in Viking zone	Viking: "fused separately from any other circuit"; 210.8(A)(6)
HD	Zephyr hood, hardwired	15 A	14/2	AFCI	J-box per listing	210.12(A); disconnect per 422.31
LT	Main + 2 pendants + under-cabinet	15 A	14/2, 14/3	AFCI	dimmers; UC on its own switch	210.70(A)(1); §150.0(k)

*GD size follows the disposal model (none selected yet).

DF = dual-function AFCI/GFCI breaker. Electrician confirms the exact Eaton catalog numbers listed for this panel. TR = tamper-resistant (406.12). Small-appliance circuits SA-A/B/C carry no lighting or other-room outlets (210.52(B)(2)).

Countertop receptacle check — 210.52(C)

Rule: no point along the counter wall more than 24 in from a receptacle; none required directly behind sink or range; boxes no more than 20 in above the counter; no face-up receptacles in the counter unless listed for it.

Wall (datum)	Counter wall space	Receptacles	Worst gap	Result
East (from SE corner, north)	0–60	E1 @ 12, E2 @ 48	12 in	PASS
South (from SE corner, west)	0–75 (corner, trash, DW top); sink 75–105 exempt; 105–115 is under 12 in wide, so none required	S1 @ 36, S2 @ 66 1/2 (+E1 covers the corner)	24 in at 12 / 8 1/2 in at 75	PASS
West (from south end, north)	0–18 and 48–66; range 18–48 exempt	W1 @ 9, W2 @ 57	9 in	PASS

Proposed height: box center 8 in above finished counter (44 in AFF) on the east and range walls; S1 at 40 in AFF under the niche; S2 at 66 1/2 in from SE, between niche and window, at 44 in AFF, inside the 20 in limit. Mark heights only after floor buildup t is known (SF mark = AFF target + t).

Lighting — Title 24 Part 6, §150.0(k)

- **Task lights over the sink run (proposal, Tisa to lock):** 3 fixtures at 18, 54 and 90 in from the SE corner — equal 36 in spacing, T3 on the new sink centerline (90 in, per Magicplan 11/2 evening) — on a line 15 in off the finished south wall. Replaces the earlier 2 pendants. See drawing KE-02.
- All fixtures JA8-marked high-efficacy; recessed cans IC-rated airtight (ICAT), no screw bases.
- Kitchen is a habitable space: **main and pendants on dimmers** (not auto-off). Pendants from entry and sink = dimmer with companion/remote, matching brand.
- **Under-cabinet lighting on its own switch**, separate from ceiling lights. Only applies if uppers are chosen.
- Wall switch at the entry controls kitchen lighting (210.70(A)(1)).

Install rules that affect rough-in

- **Routing is orthogonal — no diagonals.** Vertical runs stay in one stud bay; horizontal runs go through bored studs at set bands (east/west 50 in AFF; south 30 and 24 in AFF behind the bases); attic runs go along or across joists. Drawings KE-01 to KE-05 show every route.
- **SF 300.4(I):** wiring under 8 ft is treated as exposed to damage. Per Tisa: the DW and disposal receptacles share **one 2-gang box in the back wall of the sink base (C3), flush**, at 79 in from SE and 14 in AFF — two separate receptacles on two separate circuits, each neutral kept with its own hot, both labeled. No loose NM inside the cabinet; never behind the dishwasher (Bosch).
- Dishwasher and disposal stay on separate breakers with separate neutrals; no shared multiwire circuit.
- Hood hardwired: if the Zephyr label is over 300 VA, the breaker needs a lock-off (422.31(B)) because the panel is not in sight of the hood.
- D04 panel: 30 in wide × 36 in deep × 6.5 ft clear working space in front, kept free of furniture (110.26).
- Kitchen = Romex per Tisa. The empty-ENT hold applies to non-kitchen outlets only.
- Rough inspection before any wall or ceiling is closed.

Open items (owner left blank unless already assigned)

#	Item	Why it matters	Owner
K-E01	**S1 vs. recessed niche.** Magicplan places the niche at 28 1/16–60 1/16 in from SE, bottom 43 in AFF.	Proposal on KE-02: S1 low at 40 in AFF, under the niche bottom; niche unchanged. Owner locks.	Tisa
K-E02	**South run changed (Magicplan, 11/2 evening):** corner 0–36, trash 36–51, DW 51–75, sink base 75–105 (CL 90, under the window), filler ~7 in.	S2 moved to 66 1/2 in, between niche and window, normal height; DW/GD 2-gang box at 79 in in the sink base; no switch box at the sink (wireless remote; counter air switch or wall switch for the disposal). Cabinet sheets K002/K101 need the same update.	Kitchen cabinet owner
K-E03	**Range GFCI method.** Code requires GFCI; Viking advises against GFCI breakers and wants a GFCI receptacle on a dedicated circuit. A GFCI receptacle behind the range is not readily accessible, which code requires.	Electrician picks: DF breaker (code-clean, Viking warranty note) or AFCI breaker + dead-front GFCI in an accessible spot on the RG circuit.	Electrician
K-E04	**Zephyr hood electrical rating.** The manual on file (S11) is a 2006 image scan with no readable rating.	Sets the disconnect rule and whether the hood can share LT. Read the nameplate or current manual.	
K-E05	Disposal model not selected.	Sets GD breaker and switch rating.	Tisa

#	Item	Why it matters	Owner
K-E06	Permit not on record; contractor not named.	Without a C-10 permittee, no rough-in can start.	Tisa
K-E07	Floor buildup t unknown.	All AFF marks shift by t.	

Sequence

1. Cabinet stations marked from K002 and checked → 2. John's framing and backing done → 3. Electrician marks boxes against this plan (after K-E01 to K-E04) → 4. Permit and rough-in by the licensed contractor → 5. Rough inspection → 6. Close walls → 7. Devices, fixtures, appliance hookup, GFCI/AFCI test.

Sources

- E20 circuit register: `electrical/2026-11-01-eaton20/outputs/Circuit-and-Romex-Takeoff-REVIEW.csv`
- Kitchen Rev B K002, K101–K104, K302: `kitchen/outputs/rev-b-2026-10-29/`
- Manuals: Bosch SHV9PT53UC install (S07), Haier QJS15HYRFS (S09), Viking VGCC530 install (S10), Zephyr ZSA-E30AS (S11, unreadable scan)
- 2025 SF Electrical Code amendments — 300.4(I), 230.71, 334.10/334.12, Bldg Code 89.120, 89.122
- SF homeowner electrical permit application — single-family only
- 2025 CEC significant changes summary (Benicia) — secondary source
- 2023 NEC 210.52 summary (Mike Holt) — secondary source
- 2025 Residential Lighting Mandatory Measures (Roseville) — secondary source

Code text above comes from official SF documents and secondary summaries of the 2025 CEC. The licensed electrician and DBI inspector confirm final requirements.